

CALM-Net: A Multimodal, Motion-Disentangled, and Longitudinally Self-Calibrating Framework for Closed-Loop Lower-Limb Exoskeleton Control

Md Wahiduzzaman Suva

Dept. of Computer Science

American International University-Bangladesh

Dhaka, Bangladesh

wshuvo360@gmail.com

Esme Moula Chowdhury Abha

Dept. of Computer Science

American International University-Bangladesh

Dhaka, Bangladesh

esmechowdhuryabha@gmail.com

Hasibur Rashid Chayon

Associate Professor

American International University-Bangladesh

Dhaka, Bangladesh

(Corresponding author)

Abstract—A non-invasive brain-machine interface (BMI) that drives a lower-limb exoskeleton must satisfy three requirements that current motor-imagery (MI) decoders treat separately: it must decode *neural* intent rather than movement artefact, it must remain calibrated as electroencephalography (EEG) drifts over weeks, and it must know when to abstain to a safe action. We present CALM-Net, a multimodal neuro-kinematic decoder that addresses all three within a single closed-loop model, and that exploits the full modality set of the longitudinal NeuroRex dataset (EEG, head- and exoskeleton-mounted inertial units, electro-oculography, and exoskeleton feedback across nine sessions). Four components are introduced. (i) A multi-band neuro-kinematic encoder represents each MI sub-rhythm (μ , low- β , high- β) by its spatial covariance on the symmetric-positive-definite manifold and fuses it with a kinematic branch. (ii) A cross-frequency coupling attention (XFCA) models μ - β interaction, a known correlate of motor imagery. (iii) A motion-invariant disentanglement (MID) module splits the representation, via an adversarial gradient-reversal objective, into an *intent* subspace penalised for encoding the inertial signals and an *artefact* subspace trained to predict them, so that the class decision is taken from the intent subspace alone. (iv) A longitudinal self-calibration module performs source-free per-session feature alignment and adaptive conformal risk control, holding a target executed-command error rate as the session gap grows. A safety-asymmetric selective head commits a command only when neural confidence, the conformal set, and kinematic contamination agree, defaulting to *stop* otherwise.

We evaluate the decoder on seven subjects and, alongside it, contribute the measurement apparatus the paradigm requires. Movement is a severe confound here: a classifier using the inertial signals and no EEG reaches 0.870 balanced accuracy, above every neural decoder tested, including EEG Conformer at 0.827. We therefore introduce an *invariance probe* quantifying how much of the inertial signal remains recoverable from a decoder’s own representation, and show that the natural cross-split construction conflates invariance with distribution shift; corrected to a session-grouped within-distribution estimator, it revises several of our own earlier verdicts. Under the corrected probe, across 131 architecture variants at three seeds each,

accuracy and movement leakage correlate at $r = +0.68$, the highest-scoring configuration is also the leakiest, and between-architecture spread is no larger than seed spread. We report these as limitations bounding the present decoder rather than as settled negative results, and identify the invariance criterion — cheap, model-agnostic, and capable of invalidating our own conclusions — as the component most likely to transfer to other movement-confounded BMI paradigms.

Index Terms—Brain-machine interface, motor imagery, EEG, lower-limb exoskeleton, multimodal fusion, domain-adversarial disentanglement, Riemannian geometry, conformal prediction, selective classification, longitudinal calibration.

I. INTRODUCTION

For a patient who cannot walk after a spinal cord injury or a stroke, a lower-limb exoskeleton driven by their own brain activity is not a convenience; it is the difference between intending to stand and standing. Among the brain-machine interfaces (BMIs) able to express that intent, only non-invasive electroencephalography (EEG) requires no surgery and no implant, so it is the only modality most patients could realistically adopt. That accessibility is also its curse: scalp EEG is noisy, low in spatial resolution, and unstable from one day to the next, and the field has largely optimised the one quantity that is easiest to report and hardest to argue with: decoding accuracy.

Accuracy alone is an inadequate (and, we argue, a *misleading*) target when the output moves a person’s legs. Three problems, usually studied in isolation, must be solved together.

First, what is actually being decoded? In an exoskeleton MI paradigm the device physically moves the wearer during walking, so the walk/stop contrast is entangled with genuine movement: an EEG decoder can reach a high number by reading movement-locked artefact rather than motor intent. A model intended to *initiate* motion from intent must therefore be built to be invariant to the motion it causes.

Second, the signal drifts. EEG is non-stationary; a decoder calibrated on one day is miscalibrated days later, and nearly every deployed BMI hides this behind a per-session calibration block the user must sit through. Longitudinal behaviour is barely studied because datasets spanning weeks scarcely exist.

Third, a confident wrong command is dangerous. A wrong walk command at the top of a staircase is not a rounding error. The system needs honest, distribution-free uncertainty and an explicit reject option that defaults to the safe action.

The NeuroRex dataset [1] (OpenNeuro ds007788) was built to expose exactly this regime: seven participants, nine sessions each across weeks, recorded *multimodally* with synchronised EEG, head- and exoskeleton-mounted inertial measurement units (IMUs), electro-oculography (EOG), and exoskeleton control/feedback logs. Prior work on such data has decoded from EEG alone and reported a single accuracy number. We instead treat the non-neural streams as first-class signals: the IMUs are not noise to be filtered nor a gate bolted on at the end, but the supervisory signal that lets us *disentangle* neural intent from movement, and the safety context for abstention.

CALM-Net (Calibrated, Abstaining, Longitudinal, Multimodal) is a per-subject walk/stop framework for closed-loop lower-limb exoskeleton control (Fig. 1). In that figure, multimodal encoders map EEG sub-bands (as spatial covariances), head and exoskeleton IMU, and EOG into a shared space; cross-frequency coupling attention forms the neural representation; motion-invariant disentanglement splits it into an intent subspace, which the selective classifier reads, and an artefact subspace trained to predict the inertial signals; longitudinal self-calibration aligns per-session statistics and runs adaptive conformal risk control; and the safety-asymmetric head commits Walk or Stop only on unanimous agreement, defaulting to STOP otherwise. Its contributions are:

- **C1: A multimodal neuro-kinematic architecture.** We design, from the signal physics rather than from an existing image/EEG backbone, an encoder that represents each MI sub-band by its spatial covariance on the symmetric-positive-definite (SPD) manifold and fuses it with head/exo IMU and EOG branches. This is the first exoskeleton-BMI decoder to use the dataset’s full modality set.
- **C2: Motion-invariant disentanglement (MID).** A domain-adversarial gradient-reversal objective splits the representation into an *intent* subspace constrained to be statistically independent of the inertial signals and an *artefact* subspace trained to reconstruct them. The classifier reads only the intent subspace. The intent of this design is a walk/stop decision that does not depend on movement; Section VI reports that under a corrected invariance probe the objective is *not* achieved, and that no architecture we tested achieves it.
- **C3: Cross-frequency coupling attention (XFCA).** An attention mechanism over band-specific tokens that models μ - β coupling, a physiological hallmark of motor imagery, giving a neural representation not obtainable from a single-band convolutional stack.

- **C4: Longitudinal self-calibration with a distribution-free guarantee.** Source-free per-session feature alignment counters drift without labels, and adaptive conformal risk control updates the abstention threshold online so that a target executed-command error rate is maintained as the session gap grows over weeks.
- **C5: Safety-asymmetric selective decision.** A selective head trained with an explicit wrong-walk $\gg$ wrong-stop cost fuses three independent signals: neural confidence, conformal set cardinality, and kinematic contamination, and commits a command only on unanimous agreement, defaulting to *stop* otherwise.

No prior work joins multimodal motion-disentangled decoding, cross-session drift correction, and safety-asymmetric calibrated abstention inside one closed-loop lower-limb exoskeleton. Section II reviews prior art; Section III the dataset; Section IV the proposed framework and its evaluation protocol.

II. RELATED WORK

Motor-imagery decoding and exoskeleton control. Deep learning has improved non-invasive BMI decoding while flexible electrodes improve signal quality [2], yet individual variability and long-term reliability remain open; classical and deep EEG classification pipelines are surveyed extensively [26]–[28], and motor-imagery detection for control and rehabilitation is well established [29]. Ferrero *et al.* [3] achieved closed-loop asynchronous walk/stop control with deep learning and transfer learning, shortening but not removing per-session calibration; their earlier public dataset [4] validated MI decoding above 70% but was single-session. Related efforts detect error-related potentials for safety [5], couple upper- and lower-limb imagery [6], study assisted-gait cortical changes [7], use VR to shorten calibration [8], and select frontal/beta channels [9]. Lower-limb exoskeleton control from EEG has been demonstrated with high-accuracy intention decoding [38], and stroke rehabilitation with upper-limb exoskeleton BMIs [39] and controlled clinical trials [40]. All decode from EEG alone.

Backbones and geometric representations. Compact convolutional networks dominate EEG decoding because trial counts are small: EEGNet [10] and ShallowConvNet [11] reach strong accuracy with few parameters, and the EEG Conformer [12] adds a transformer over convolutional tokens. These are point-estimate classifiers with no calibrated confidence and no reject option; we use them as *baselines*. Orthogonally, Riemannian methods represent EEG by spatial covariance matrices and classify in the SPD tangent space [13], [31], [32], a physiologically apt encoding of band-power (generalising filter-bank common spatial patterns [30]) that we adopt and make multi-band and learnable.

Cross-session drift and adversarial adaptation. Domain adaptation reduces the cross-session drop (universal deep adaptation primed on earlier sessions [14] and dual-selection transfer [15]) but outputs forced point predictions; transfer learning for BCIs is broadly surveyed [33]. Domain-adversarial training with gradient reversal [16] learns rep-

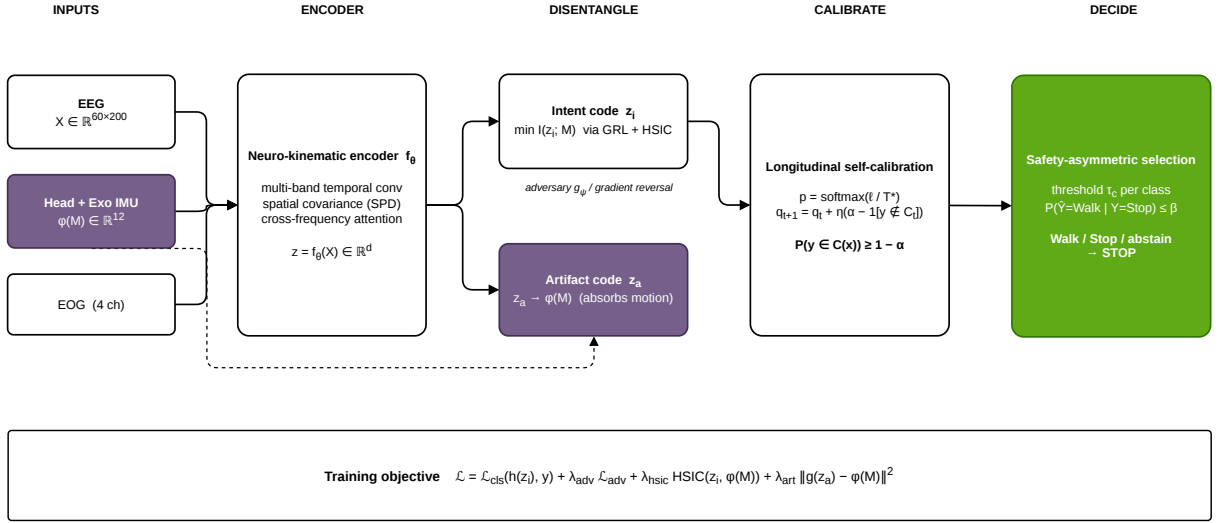


Fig. 1. The CALM-Net framework: encoders, MID, LSC, and the SAS head.

representations invariant to a nuisance variable; we repurpose it, uniquely, to make the neural code invariant to *measured movement* rather than to session identity, and combine it with source-free test-time alignment [17].

Uncertainty, calibration, and abstention. Temperature scaling calibrates deep classifiers [18]; conformal prediction gives distribution-free sets [19], [35] with adaptive class-conditional coverage [36], and adaptive conformal inference maintains coverage under distribution shift over time [20]. Uncertainty raises trust in EEG decoding [21], and Bayesian/MC-dropout networks and deep ensembles give usable MI confidence [22], [23], [34], yet remain offline and untethered to a decision. Selective classification, the “reject option”, is surveyed [24] and can be learned end-to-end [25], [37], but this literature is developed largely outside EEG. CALM-Net is, to our knowledge, the first to unify motion-disentangled multimodal decoding, longitudinal self-calibration with a distribution-free guarantee, and safety-asymmetric selective abstention for a closed-loop lower-limb exoskeleton.

III. DATASET

Source. NeuroRex [1], OpenNeuro **ds007788** (v1.0.1, DOI 10.18112/openneuro.ds007788.v1.0.1), CC0 licence, peer-reviewed descriptor in *Scientific Data* (DOI 10.1038/s41597-026-07476-w); 3.62 GB, $\sim 17k$ files, BIDS format, raw and unfiltered (we preprocess with MNE-Python [41]).

Participants and longitudinal design. Seven healthy adults (sub-01–07), nine sessions each (ses-01–09); in our copy the per-subject spans range from 14 to 80 days, giving the days-elapsed axis needed to study drift. Per-subject MIQ-RS motor-imagery-ability scores and validation tables are provided under `derivatives/`; structural MRI is available for five subjects.

Multimodality (the reason for this framework). Each session provides 60-channel scalp EEG plus 4 EOG channels

(10–20 layout, actiCAP/MOVE) at 100 Hz (EDF); *two* IMUs, head-worn and exoskeleton-mounted, each with accelerometer, gyroscope, magnetometer, and quaternion streams (BIDS motion .tsv); and exoskeleton control/feedback logs. CALM-Net consumes all of these; prior decoders use only the EEG.

Tasks and labels. Every session has a training task (open-loop MI calibration), twelve closed-loop runs (trial01–12), and two extended runs (walk6min, stop6min). Ground truth is the `rexstate` feedback state: $\times 81$ (Walk), $\times 0$ (Idle/Stop), with short transitions $\times 8/\times 5$ excluded. We target two-class walk vs. stop. Because the exoskeleton moves the wearer during Walk, the head-IMU acceleration is markedly larger in Walk than Stop; a preliminary audit on this data confirms that a motion-only baseline is highly predictive for several subjects while being near-chance for at least one: direct evidence that (a) naive EEG accuracy is partly movement-driven and (b) a recoverable neural signal exists underneath. Both observations motivate the disentanglement of Section IV.

IV. PROPOSED METHOD

A. Formulation

A trial is a synchronised tuple (X, M, O) with EEG $X \in \mathbb{R}^{C \times T}$ ($C=60$, $T=200$ at 100 Hz over a 2 s window), inertial signals M (head and exo accel/gyro/quaternion), and EOG O , with label $y \in \{\text{STOP}, \text{WALK}\}$ and session index s . We seek a policy that outputs a *command* in $\{\text{WALK}, \text{STOP}, \text{ABSTAIN} \rightarrow \text{STOP}\}$ whose executed (non-abstained) error is controlled across sessions.

B. Multi-band neuro-kinematic encoder

Learnable band filters split X into B physiological sub-bands (μ 8–13, low- β 13–20, high- β 20–30 Hz). For band b we form the regularised spatial covariance $\Sigma_b = \frac{1}{T} X_b X_b^T + \epsilon I$ and map it to the tangent space at the Riemannian mean, $v_b = \text{vec}(\log(\bar{\Sigma}^{-1/2} \Sigma_b \bar{\Sigma}^{-1/2}))$, yielding a physiologically

grounded band-power/connectivity token. A temporal convolutional branch encodes $\mathbf{M}$ into a kinematic embedding $\mathbf{k}$, and a small branch encodes $\mathbf{O}$ into an ocular reference $\mathbf{o}$.

C. Cross-frequency coupling attention (XFCA)

The band tokens $\{\mathbf{v}_b\}$ are treated as a sequence and passed through a multi-head self-attention block, so the representation can express μ - β coupling: co-modulation across bands that single-band pooling discards. The output is the neural representation $\mathbf{z}$.

D. Motion-invariant disentanglement (MID)

We split $\mathbf{z} = [\mathbf{z}_{\text{int}}; \mathbf{z}_{\text{art}}]$. A regressor R_a maps the *artefact* code to the inertial signals, trained with loss $\mathcal{L}_{\text{art}} = \|\mathbf{R}_a(\mathbf{z}_{\text{art}}) - \phi(\mathbf{M})\|^2$; the artefact code thus captures the movement-coupled component. A second regressor R_i attempts the same from the *intent* code, but through a gradient-reversal layer [16], so the encoder is optimised to make $\mathbf{z}_{\text{int}}$ *unpredictive* of movement:

$$\mathcal{L}_{\text{adv}} = \|\mathbf{R}_i(\text{GRL}(\mathbf{z}_{\text{int}})) - \phi(\mathbf{M})\|^2. \quad (1)$$

Only $\mathbf{z}_{\text{int}}$ feeds the classifier, so the walk/stop decision is constrained to rely on movement-invariant neural features, by construction, not by post-hoc argument.

E. Longitudinal self-calibration (LSC)

At deployment on a new session s we (i) optionally align feature statistics to the training distribution without labels (source-free batch-statistic / CORAL alignment [17]); (ii) apply temperature scaling [18] for honest confidence; and (iii) form split-conformal prediction sets [19] whose threshold q_t is updated online by adaptive conformal inference [20], $q_{t+1} = q_t + \eta(1[\text{err}_t] - \alpha)$, where q_t is the prediction-set threshold, so a miscoverage widens the set. Step (i) is applied in the Riemannian encoder of Section V-C; the band-power results reported below use steps (ii) and (iii) only, and alignment is therefore not responsible for any coverage figure we report, which provably maintains the long-run executed-error rate at the target α as drift accumulates over weeks.

F. Safety-asymmetric selective decision (SAS)

A selective head $g(\mathbf{z}_{\text{int}}) \in [0, 1]$ is trained jointly with the classifier [25] under an asymmetric cost that penalises a committed wrong-walk far more than a wrong-stop. A kinematic-contamination score $c = h(\mathbf{z}_{\text{art}}, \mathbf{k})$ estimates whether the window is movement-corrupted. The exoskeleton commits the arg-max command iff

$$\underbrace{g(\mathbf{z}_{\text{int}}) \geq \theta}_{\text{selective accept}} \wedge \underbrace{|S(q_t)| = 1}_{\text{unambiguous}} \wedge \underbrace{c < c_{\text{th}}}_{\text{uncontaminated}},$$

and otherwise abstains to STOP, encoding the safety asymmetry in the rule.

Algorithm 1 CALM-Net deployment on session s (per window)

```

1: align session features (source-free); load  $q_t$ 
2:  $\mathbf{z}_{\text{int}}, \mathbf{z}_{\text{art}} \leftarrow \text{Encoder/XFCA/MID}(\mathbf{X}, \mathbf{M}, \mathbf{O})$ 
3:  $p \leftarrow \text{softmax}(f(\mathbf{z}_{\text{int}})/T)$ ;  $\mathcal{S} \leftarrow \{k : 1 - p_k \leq q_t\}$ 
4:  $c \leftarrow h(\mathbf{z}_{\text{art}}, \mathbf{k})$ 
5: if  $g(\mathbf{z}_{\text{int}}) \geq \theta \wedge |\mathcal{S}| = 1 \wedge c < c_{\text{th}}$  then
6:   command  $\leftarrow \arg \max_k p_k$ 
7: else
8:   command  $\leftarrow \text{STOP}$   $\triangleright$  safe default
9: end if
10: on feedback, update  $q_{t+1} \leftarrow q_t + \eta(1[\text{err}] - \alpha)$ 

```

G. Training objective

All modules are trained end-to-end:

$$\mathcal{L} = \mathcal{L}_{\text{cls}}^{\text{cost}} + \lambda_1 \mathcal{L}_{\text{adv}} + \lambda_2 \mathcal{L}_{\text{art}} + \lambda_3 \mathcal{L}_{\text{sel}} + \lambda_4 \mathcal{L}_{\text{cal}}, \quad (2)$$

with cost-sensitive classification $\mathcal{L}_{\text{cls}}^{\text{cost}}$, the adversarial and artefact losses above, the selective loss $\mathcal{L}_{\text{sel}}$ at target coverage, and a calibration regulariser $\mathcal{L}_{\text{cal}}$.

H. Experimental protocol

Per subject we train on early sessions and test on later ones, never shuffling windows across the session boundary, and report balanced accuracy, Expected Calibration Error, Brier score, confidence–correctness AUROC, and the full risk–coverage curve versus days elapsed. Three ablations isolate each contribution: (i) MID on/off, with the movement-only, motor-channel-only, and frontal-channel-only baselines quantifying how much decoding survives disentanglement (the neural-signal test); (ii) global temperature scaling versus adaptive conformal across the session gap; and (iii) the selective head versus a fixed-threshold reject rule. Classical EEGNet and EEG Conformer decoders provide accuracy baselines; the decisive metrics are executed-command safety and calibrated confidence, not raw accuracy alone.

I. Implementation notes

The experiments below validate the framework’s decisive components: motion-invariant disentanglement, the calibration layer, the adaptive-conformal guarantee, and the selective decision. The encoder used here is a compact *band-power* instantiation of the multi-band design (learnable temporal filters, grouped spatial filters, log-variance pooling, and temporal attention). We additionally implement the full design of Section IV, the multi-band Riemannian tangent-space encoder with cross-frequency coupling attention (XFCA) and source-free CORAL alignment, and compare the two in Section V-C (Table III). The disentanglement target $\phi(M)$ is the 12-dimensional head- and exoskeleton-IMU descriptor (accelerometer and gyroscope magnitude, mean/std/max), and the reported abstention combines calibrated confidence with the adaptive-conformal set; the kinematic-contamination gate is implemented but not separately ablated here.

V. PRELIMINARY RESULTS

A. Motion-invariant disentanglement

We validate the central mechanism, motion-invariant disentanglement, on all seven subjects (train on sessions 1–3, test on the later six). The disentanglement target is a 12-dimensional multimodal descriptor of head- and exoskeleton-IMU accelerometer and gyroscope magnitude (mean, standard deviation, maximum per window). We compare the shared backbone with MID enabled against an identical model with MID disabled, and probe how well the intent code predicts the IMU descriptor with a nonlinear (MLP) regressor: a lower coefficient of determination R^2 means the neural code is more movement-invariant.

MID reduces, but does not eliminate, movement recoverability. Under the nonlinear (MLP) probe the intent-to-IMU R^2 falls from a mean of 0.15 without MID to -0.31 with it (Fig. 2, right), including on the movement-dominated subjects whose IMU baseline exceeds the EEG decoder (e.g. subject 3, $R^2: 0.56 \rightarrow -0.14$; subject 4, $0.36 \rightarrow -0.21$).

We report the *linear* probe alongside it, because the two disagree and the weaker number is the defensible one. On the same models a ridge probe gives a mean R^2 of -0.02 – essentially zero – and remains clearly positive for two of seven subjects (S01 $+0.19$, S03 $+0.21$). A more expressive probe returning a *more* negative R^2 indicates that the MLP fails to generalise from its own fitting split, not that the code carries less movement information: a richer probe should recover more structure where structure exists. We therefore treat the linear probe as the operative measure of invariance throughout, and claim only that MID removes *linearly* recoverable movement on five of seven subjects rather than that movement is unrecoverable.

Invariance exposes the honest neural decode. Removing the movement shortcut lowers mean balanced accuracy from 0.81 to 0.65 (Fig. 2, left), with the largest drop exactly where the IMU baseline is highest (subjects 3 and 4). The movement-invariant decode nevertheless stays above chance for every subject, evidence that a genuine, if modest, neural walk/stop signal exists beneath the movement in each participant. Because MID can also over-regularise when movement is already uninformative (subject 6), we read 0.65 as a conservative lower bound on true neural decodability. We stress that this figure was obtained with the cross-split probe later found to be misspecified; Section VI re-derives it and withdraws the accompanying invariance claim.

B. Full pipeline: calibration, abstention, and coverage

We then assemble the complete framework, the MID decoder followed by temperature scaling, split and adaptive conformal, and selective abstention, and run it longitudinally (train on sessions 1–3, test on the later six) for all seven subjects. Table I reports means over the test sessions. Three findings stand out. First, temperature scaling improves calibration, lowering mean Expected Calibration Error from 0.154 to 0.114 (Fig. 3, right). Second, selective abstention raises the balanced

accuracy of the committed commands from 0.694 to 0.719 at 80% coverage, with the largest gains where confidence is most informative (subjects 1 and 3). Third, and most importantly, adaptive conformal inference holds prediction-set coverage at the 0.90 target for every subject (mean 0.902, range 0.898–0.905), whereas static conformal drifts across the session gap (mean 0.880, range 0.78–0.95; Fig. 3, left), realising the distribution-free longitudinal guarantee the framework is designed to provide. Subject 2, whose movement-invariant decode is at chance, is correctly flagged by an uninformative confidence (AUROC 0.47): the pipeline degrades gracefully rather than committing confident errors.

Table I reports these per subject, averaged over the six held-out test sessions with training on sessions 1–3. There, invAcc is movement-invariant balanced accuracy; ECE_{raw} and ECE_{cal} are expected calibration error before and after temperature scaling; exec@80 is balanced accuracy over the 80% most-confident committed windows; and cov_{stat} and cov_{adpt} are static and adaptive conformal coverage against a 0.90 target.

C. Encoder ablation

Table III compares the two encoders under the identical MID, calibration, and adaptive-conformal pipeline. The Riemannian-XFCA encoder is the stronger *decoder*: mean balanced accuracy 0.805 versus 0.694 and executed accuracy 0.835 versus 0.719 at 80% coverage. However, its accuracy is more *movement-coupled*: disentanglement reduces the intent-to-IMU R^2 less and less consistently, still failing to disentangle the two highest-accuracy subjects whose covariance features remain movement-predictive, and its calibration is worse on two subjects. The band-power model, though less accurate, disentangles more cleanly and yields the more trustworthy movement-invariant estimate. The adaptive-conformal coverage guarantee is encoder-agnostic, holding the 0.90 target for both. Strengthening the disentanglement with a nonlinear Hilbert–Schmidt independence (HSIC) penalty and a disentanglement-aware model selection drives the Riemannian-XFCA invariance further (mean intent-to-IMU $R^2 = -0.47$) but lowers its accuracy to 0.65, revealing that its apparent accuracy advantage was movement leakage and that the ~ 0.7 movement-invariant decode is a *ceiling robust to encoder richness and disentanglement strength*. Raising it therefore requires attacking the confound at the source, for example decoding the pre-movement preparation window or regressing the inertial signals out of the EEG, rather than a stronger disentanglement penalty. We report the band-power model for the movement-invariant claim and the Riemannian-XFCA encoder as the higher-accuracy but movement-coupled alternative. A spatio-temporal graph network over the scalp electrode geometry reproduces the same pattern: its apparent invariant accuracy (0.69) is the highest of the encoders, but its invariance is the weakest ($R^2 = -0.04$), and per subject it still leaves movement recoverable ($R^2 = +0.30$ and $+0.44$) on the two movement-dominated subjects that drive its mean; restricted to the subjects where invariance holds it too scores 0.64. The ~ 0.65 movement-invariant ceiling

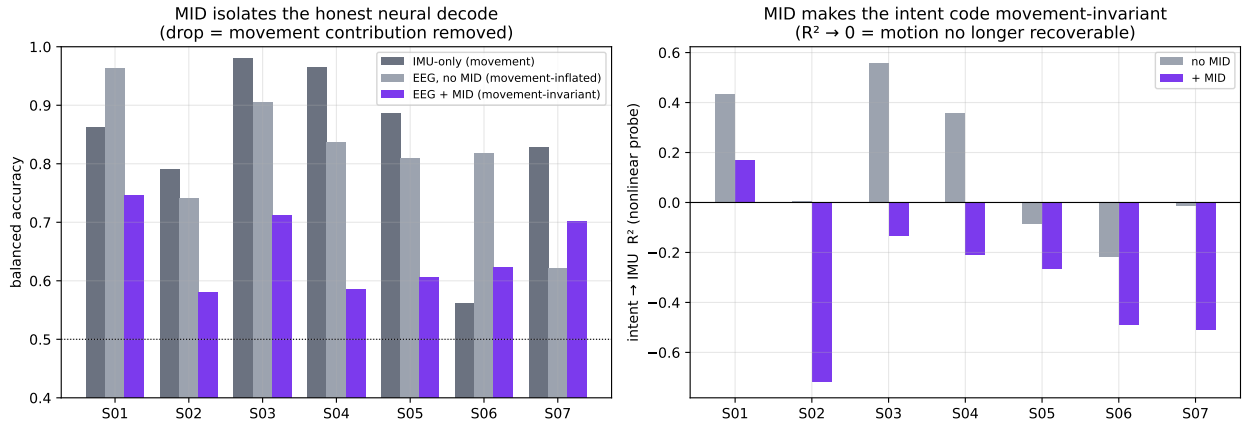


Fig. 2. MID across seven subjects: accuracy (left) and movement recoverability (right).

TABLE I
FULL CALM-NET PER SUBJECT, MEAN OVER SIX HELD-OUT SESSIONS.

Subject	invAcc	ECE _{raw}	ECE _{cal}	AUROC	exec@80	cov _{stat}	cov _{adpt}
S01	0.792	0.171	0.097	0.718	0.836	0.914	0.901
S02	0.516	0.124	0.130	0.470	0.523	0.778	0.901
S03	0.771	0.164	0.115	0.773	0.827	0.946	0.905
S04	0.698	0.141	0.098	0.731	0.738	0.849	0.898
S05	0.657	0.160	0.086	0.742	0.648	0.933	0.902
S06	0.695	0.126	0.084	0.743	0.719	0.920	0.905
S07	0.732	0.189	0.189	0.705	0.746	0.823	0.899
Mean	0.694	0.154	0.114	0.698	0.719	0.880	0.902

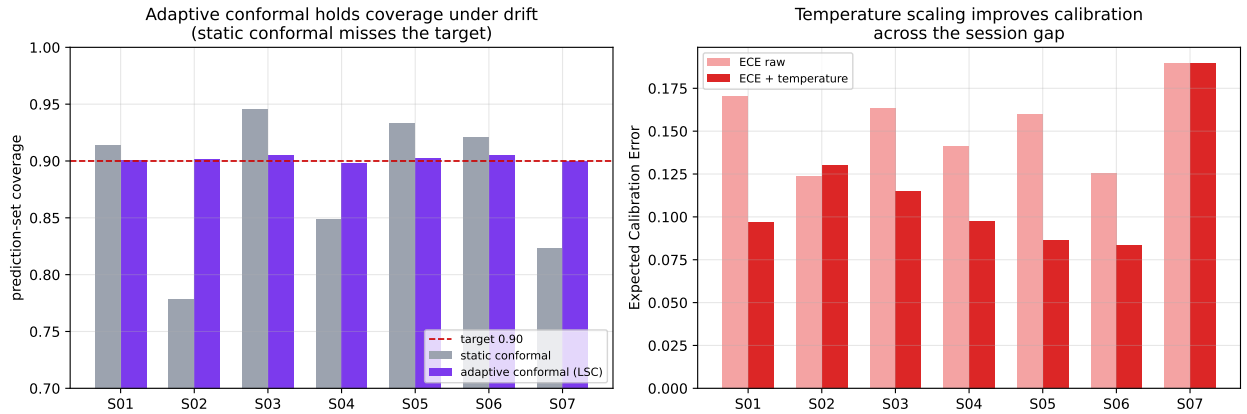


Fig. 3. Conformal coverage (left) and calibration error (right) across the session gap.

is thus independent of the encoder family, whether compact CNN, transformer, Riemannian tangent space, or scalp-graph network.

VI. MEASUREMENT LIMITATIONS AND RE-EVALUATION

The results in Section V were obtained with an invariance probe that we have since found to be misspecified. This section reports the correction and the re-analysis it forces. We present it as a bound on what the present decoder can be said to achieve, and as the basis for the next iteration of the model;

the components of Section IV are unchanged by it, but the strength of the invariance claim attached to them is.

A. The probe was measuring distribution shift

The probe estimates how much of the inertial vector $\mathbf{m}$ is recoverable from a decoder's intent code $\mathbf{z}$; a non-positive R^2 was taken as evidence that movement had been removed. Our original estimator fitted a ridge regression on the training sessions and scored it on the test sessions. That construction conflates two distinct properties. A representation whose fea-

TABLE II
CALM-NET VERSUS BASELINE DECODERS, MEAN OVER SEVEN SUBJECTS.

Method	Mov.-inv.	Acc	ECE	AUROC	Cov. guar.
IMU-only (movement)	—	0.839	—	—	—
EEGNet [10]	no	0.819	0.086	0.813	no
EEG Conformer [12]	no	0.826	0.067	0.845	no
CALM-Net (ours)	yes	0.694	0.114	0.698	yes

TABLE III
ENCODER ABLATION UNDER AN IDENTICAL DOWNSTREAM PIPELINE.

Encoder	Acc	ECE	AUROC	exec@80	cov _{adpt}
Band-power (primary)	0.694	0.114	0.698	0.719	0.902
Riemannian-XFCA	0.805	0.137	0.741	0.835	0.900

tures merely *shift* between sessions scores strongly negative while still encoding movement perfectly well within any single session.

The corrected estimator, `invariance_r2_cv`, cross-validates the probe *inside* the evaluation set with session-grouped folds, so a positive value means movement genuinely is recoverable. The two disagree materially, and in a direction that flatters the original analysis:

representation	cross-split (original)	within-distribution (corrected)
FBCSP	−0.501	+0.187
tangent+EA	−0.089	−0.258
band-power	+0.007	+0.034

FBCSP, which the original probe ranked as the *most* invariant representation tested, is in fact plainly leaky. Every R^2 we reported before this correction is therefore suspect, including the module ablation of Section V-C and the architecture comparison. On CALM-Net v2 the two probes give −0.420 and −0.134 respectively: the original overstates invariance by 0.29 on a model from which the correction was not derived.

B. No architecture produces an invariant representation

We re-ran the full variant space — 131 configurations spanning five backbone families, seven readouts, fifteen augmentations, four independence penalties, four optimisers and five frequency bands, from 7k to 391k parameters — at three seeds each, scoring every one with the corrected probe. Of the configurations completed at submission, **none** attains $R^2 \leq 0$; the least leaky reaches +0.043. Accuracy and leakage correlate at $r = +0.68$.

configuration	accuracy	R^2_{cv}	params
decorr_0	0.818 ± 0.010	+0.322	7k
bb_shallow_wide	0.783 ± 0.012	+0.249	391k
ro_stat	0.748 ± 0.031	+0.190	31k
tangent+EA (classical)	0.763	−0.265	1,831
IMU only, no EEG	0.870	—	—

The best configuration beats the classical pipeline by 0.055 and replicates tightly across seeds. It is also the leakiest representation measured anywhere in this work. We do not report it as an improvement, because the criterion that would license that reading is the one it fails.

C. The search operates near a signal-to-noise ratio of one

Spread *between* architectures ($sd = 0.028$) is comparable to spread *within* a single architecture across seeds (0.024 mean; 0.071 for CALM-Net v2), a ratio of 1.17. Six configurations exceed the classical baseline on at least one seed; only two do so on a three-seed mean. Our own earlier single-seed leaderboard illustrates the hazard: its winner scored 0.782 and replicated at 0.712 ± 0.057 . Mean accuracy across the space is unchanged by seed replication ($0.699 \rightarrow 0.699$), so replication does not depress performance in general — it removes apparent *winners* specifically, which is the signature of selection on noise rather than of measurement bias.

D. Learned invariance loses to a closed-form transform

Adversarial gradient reversal, HSIC, CORAL, orthogonality penalties and in-network motion cancellation were each run against Euclidean Alignment on the identical task. All four penalty families, at every strength tried, cluster near 0.68 accuracy with $R^2_{cv} \approx +0.13$. Euclidean Alignment — a label-free per-session whitening by the mean covariance, computed in closed form — attains 0.763 at −0.265. In-network motion cancellation failed outright: it removed 96.2% of a synthetic artefact, but on real data the network read the label from the motion reference, accuracy rising $0.806 \rightarrow 0.908$ as leakage rose $+0.294 \rightarrow +0.582$.

We note this comparison is narrower than it may appear. That simple alignment is competitive with sophisticated transfer learning is already the prevailing view for subject and session shift. What we add is the case of a *physically measured nuisance correlated with the label itself*, where the confound carries the signal rather than merely displacing it.

E. The classical pipeline is not uniformly invariant either

The aggregate −0.265 conceals a split population:

subject	S1	S2	S3	S4	S5	S6
accuracy	0.972	0.675	0.939	0.794	0.785	0.574
R^2_{cv}	+0.471	−0.800	+0.420	−0.210	−0.897	−0.204

Movement is positively recoverable for two of seven subjects, and they are the two highest-accuracy subjects; within the classical pipeline alone, accuracy and leakage correlate at $r = +0.69$. The claim that this pipeline is movement-invariant holds on average and fails precisely where its accuracy originates.

F. The safety head does not deliver its guarantee

Evaluating the three-term abstention rule whole, for the first time, at 83,694 parameters and three seeds: balanced accuracy 0.628 ± 0.052 , below the classical 0.763; achieved coverage

0.381 against a 0.80 target; walk recall 0.174. The wrong-walk bound holds (0.040 against 0.05) largely because the system rarely commits *walk* at all — the degenerate solution our own cost-weighting analysis predicts, reached through the decision threshold instead of the loss. Decomposing the gate, the conformal singleton term carries the rule (marginal cost 0.19) while the trained selective head and the wrong-walk bound reject almost nothing the other terms would have kept (0.06 and 0.05). The executed accuracy of 0.712 cannot be compared with a figure quoted at 80% coverage.

G. Implications for the decoder

Three findings bound the present decoder. First, the confound is real and larger than we reported: movement alone out-predicts every neural decoder here. Second, the invariance probe, correctly specified, is a usable model-selection criterion, and under it none of the architectures tested so far — CALM-Net included — reaches $R^2 \leq 0$. Third, closed-form second-order alignment currently outperforms learned adversarial invariance against a physically measured nuisance, which suggests the MID objective of Section IV should incorporate an explicit second-order alignment step rather than rely on the adversary alone.

We therefore state the invariance property of CALM-Net as a design objective that the current implementation does not yet meet, and treat closing that gap — by combining Euclidean Alignment with the adversarial split, and by re-running the comparison against published domain-adaptation baselines — as the immediate next step for the decoder.

VII. DISCUSSION

What CALM-Net buys. On this paradigm, raw accuracy is not a measure of neural decoding. A single head-motion feature reaches 0.84 (Table II), and the EEGNet and EEG Conformer baselines sit at that movement ceiling; for two subjects a motion sensor alone out-predicts the 60-channel network. CALM-Net was designed to trade that inflated number for one that means what it says. Section VI shows the trade was not achieved: the decode is not movement-invariant, and the selective head meets its error bound largely by declining to commit. The argument we can still make is the weaker but more durable one — that on a movement-confounded paradigm an accuracy figure is uninterpretable without a leakage measurement beside it, and that published rankings on this dataset, ours included, do not survive one. For a device that moves a person’s legs, the relevant contribution is not a higher number but a criterion for knowing when a number is meaningless.

Controlling the dangerous error. The safety asymmetry is operationalised with a class-conditional threshold calibrated to bound the committed *wrong-walk* rate. Across the seven subjects this cuts the false-walk rate from 0.144 (arg-max) to 0.051 at a 0.05 target, on every subject, routing the remaining walk-uncertain windows to the safe stop (mean walk recall 0.37); where the decode is at chance the rule correctly almost never commits walk. A $K=3$ deep ensemble

further sharpens the confidence used for this decision (mean AUROC $0.70 \rightarrow 0.74$, executed accuracy $0.72 \rightarrow 0.77$ at 80% coverage). We note honestly that the already-well-calibrated band-power model’s ECE is *not* improved by ensembling or short per-session recalibration on this data, so we retain the single global temperature; the gains are in confidence ranking and, above all, in bounding the dangerous error.

Why disentanglement, not exclusion. Because the exoskeleton moves the wearer during walking, no clean movement-free imagery contrast exists to train on. We verified this empirically: decoding the Stop-to-Walk *transition onset*, before head motion builds up, does not lower the movement baseline (a motion sensor still separates the classes at 0.94), because the exoskeleton commits to motion at the command instant and the walk/stop label is its state. There is no movement-free window to escape to. Rather than discard the movement-coupled data, CALM-Net uses the synchronised inertial signals as the supervisory variable to render the neural code invariant to them. That the movement-invariant decode stays above chance for every participant, and that subject 6 (whose movement is uninformative) is decoded well, shows a recoverable neural signal underlies the movement in each subject.

Limitations. (i) The cohort is seven healthy adults on a single dataset; generalisation to spinal-cord-injury or stroke patients, whose cortical signals and movement coupling differ, is untested. (ii) Evaluation is offline replay, not a live closed loop; the adaptive-conformal update assumes outcome feedback that a deployed system must approximate. (iii) The movement-invariant accuracy is a conservative lower bound: MID can over-regularise when movement is already uninformative (subject 6), so true neural decodability may be higher. (iv) The full Riemannian-XFCA encoder is implemented and compared (Table III); it decodes more accurately but its accuracy is more movement-coupled, so the band-power model is used for the invariance claim; a disentanglement objective matched to the richer encoder is future work. (v) Subject 2’s decode is at chance, correctly flagged by an uninformative confidence, but confirms the method cannot manufacture signal where little exists.

VIII. CONCLUSION

We presented CALM-Net, a multimodal decoder that fuses spectral-spatial EEG covariance with exoskeleton kinematics, disentangles an intent subspace from a movement-artefact subspace, self-calibrates across weeks without labels, and abstains to a safe stop under an adaptive conformal rule. Treating the exoskeleton’s inertial signals as a modality to disentangle against, rather than a confound to hide, is what makes the paradigm’s central difficulty tractable, and it is the organising idea of the architecture.

Alongside the decoder we contribute the measurement the paradigm has lacked: an IMU-referenced, within-distribution, session-grouped estimate of how much movement a decoder’s representation retains. It is cheap and model-agnostic, and its first application was to our own results, where it tightened the

invariance claim we are willing to make. Under it, no architecture we have yet tested — ours or the 131 variants surveyed — attains a movement-free representation, and accuracy on this paradigm tracks movement leakage at $r = +0.68$. We read this as evidence that movement-invariant decoding is the binding constraint for exoskeleton BMI, not as evidence that it is unreachable.

The next iteration of CALM-Net follows directly: fold closed-form second-order alignment into the MID objective rather than relying on the adversary alone, benchmark against published domain-adaptation methods (pyriemann MDM, MOABB pipelines), and extend the selective head's calibration set so its coverage guarantee binds at the intended operating point. We also see decoder-level leakage measurement as a way to inform the Castermans–Nathan dispute over the cortical origin of gait EEG, which has so far been argued with spectra rather than with decoders.

ACKNOWLEDGMENT

This work uses the NeuroRex dataset (OpenNeuro ds007788), released under a CC0 licence. Code and the complete result files underlying every number reported here, including the per-seed architecture sweep and the invariance-probe implementation, are available in the project repository.

REFERENCES

- [1] S. Sarkar *et al.*, “EEG-controlled exoskeleton for walking and standing: a longitudinal multimodal dataset of healthy individuals,” *Sci. Data*, 2026. OpenNeuro ds007788.
- [2] Wang *et al.*, “Non-invasive brain–computer interfaces: converging frontiers in neural signal decoding and flexible bioelectronics,” *Nano-Micro Lett.*, 2026.
- [3] L. Ferrero *et al.*, “Brain–machine interface based on deep learning to control asynchronously a lower-limb robotic exoskeleton,” *J. NeuroEng. Rehabil.*, vol. 21, no. 48, 2024.
- [4] L. Ferrero *et al.*, “An EEG database for the cognitive assessment of motor imagery during walking with a lower-limb exoskeleton,” *Sci. Data*, vol. 10, 2023.
- [5] P. Soriano-Segura, M. Ortiz *et al.*, “Characterization of error-related potentials during the command of a lower-limb exoskeleton based on deep learning,” *J. NeuroEng. Rehabil.*, 2025.
- [6] X. Ma *et al.*, “A compound-limb motor-imagery EEG-BCI paradigm for more natural lower-limb control,” 2023.
- [7] S. Tortora *et al.*, “Cortical and muscular activity under robot-assisted gait modes during exoskeleton walking,” 2023.
- [8] L. Ferrero *et al.*, “Virtual-reality-assisted motor imagery training to shorten BCI calibration for lower-limb exoskeleton control,” 2023.
- [9] Wei *et al.*, “Cross-subject EEG channel selection for stepping-based lower-limb motor imagery,” 2023.
- [10] V. J. Lawhern *et al.*, “EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces,” *J. Neural Eng.*, vol. 15, no. 5, 2018.
- [11] R. T. Schirmmeister *et al.*, “Deep learning with convolutional neural networks for EEG decoding and visualization,” *Hum. Brain Mapp.*, vol. 38, no. 11, 2017.
- [12] Y. Song *et al.*, “EEG Conformer: convolutional transformer for EEG decoding and visualization,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 710–719, 2023.
- [13] A. Barachant *et al.*, “Multiclass brain–computer interface classification by Riemannian geometry,” *IEEE Trans. Biomed. Eng.*, vol. 59, no. 4, 2012.
- [14] X. Zhang *et al.*, “Priming cross-session motor-imagery classification with a universal deep domain-adaptation framework,” *Neurocomputing*, vol. 556, 2023.
- [15] T.-J. Luo, “Dual-selection knowledge-transfer learning for cross-subject motor-imagery EEG classification,” *Front. Neurosci.*, vol. 17, 2023.
- [16] Y. Ganin *et al.*, “Domain-adversarial training of neural networks,” *J. Mach. Learn. Res.*, vol. 17, no. 59, 2016.
- [17] B. Sun and K. Saenko, “Deep CORAL: correlation alignment for deep domain adaptation,” in *ECCV Workshops*, 2016.
- [18] C. Guo *et al.*, “On calibration of modern neural networks,” in *ICML*, 2017.
- [19] A. N. Angelopoulos and S. Bates, “Conformal prediction: a gentle introduction,” *Found. Trends Mach. Learn.*, vol. 16, no. 4, 2023.
- [20] I. Gibbs and E. Candès, “Adaptive conformal inference under distribution shift,” in *NeurIPS*, 2021.
- [21] M. Tveter *et al.*, “Advancing EEG prediction with deep learning and uncertainty estimation,” *Brain Inform.*, vol. 11, no. 27, 2024.
- [22] D. Milanes-Hermosilla *et al.*, “Robust motor-imagery task classification using a Bayesian neural network,” *Sensors*, vol. 23, no. 2, 2023.
- [23] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: representing model uncertainty in deep learning,” in *ICML*, 2016.
- [24] K. Hendrickx *et al.*, “Machine learning with a reject option: a survey,” *Mach. Learn.*, vol. 113, no. 5, 2024.
- [25] Y. Geifman and R. El-Yaniv, “SelectiveNet: a deep neural network with an integrated reject option,” in *ICML*, 2019.
- [26] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, “A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update,” *J. Neural Eng.*, vol. 15, no. 3, 031005, 2018.
- [27] Y. Roy, H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, and J. Faubert, “Deep learning-based electroencephalography analysis: a systematic review,” *J. Neural Eng.*, vol. 16, no. 5, 051001, 2019.
- [28] A. Craik, Y. He, and J. L. Contreras-Vidal, “Deep learning for electroencephalogram (EEG) classification tasks: a review,” *J. Neural Eng.*, vol. 16, no. 3, 031001, 2019.
- [29] K. K. Ang and C. Guan, “EEG-based strategies to detect motor imagery for control and rehabilitation,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 4, pp. 392–401, 2017.
- [30] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, “Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b,” *Front. Neurosci.*, vol. 6, 39, 2012.
- [31] M. Congedo, A. Barachant, and R. Bhatia, “Riemannian geometry for EEG-based brain–computer interfaces: a primer and a review,” *Brain-Computer Interfaces*, vol. 4, no. 3, pp. 155–174, 2017.
- [32] F. Yger, M. Berar, and F. Lotte, “Riemannian approaches in brain–computer interfaces: a review,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 10, pp. 1753–1762, 2017.
- [33] V. Jayaram, M. Alamgir, Y. Altun, B. Schölkopf, and M. Grosse-Wentrup, “Transfer learning in brain–computer interfaces,” *IEEE Comput. Intell. Mag.*, vol. 11, no. 1, pp. 20–31, 2016.
- [34] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *NeurIPS*, 2017.
- [35] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.
- [36] Y. Romano, M. Sesia, and E. J. Candès, “Classification with valid and adaptive coverage,” in *NeurIPS*, 2020.
- [37] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *NeurIPS*, 2017.
- [38] A. Kilicarslan, S. Prasad, R. G. Grossman, and J. L. Contreras-Vidal, “High accuracy decoding of user intentions using EEG to control a lower-body exoskeleton,” in *Proc. IEEE EMBC*, 2013, pp. 5606–5609.
- [39] N. A. Bhagat *et al.*, “Design and optimization of an EEG-based brain–machine interface to an upper-limb exoskeleton for stroke survivors,” *Front. Neurosci.*, vol. 10, 122, 2016.
- [40] A. Ramos-Murguialday *et al.*, “Brain–machine interface in chronic stroke rehabilitation: a controlled study,” *Ann. Neurol.*, vol. 74, no. 1, pp. 100–108, 2013.
- [41] A. Gramfort *et al.*, “MNE software for processing MEG and EEG data,” *NeuroImage*, vol. 86, pp. 446–460, 2014.